

Omkar Sudhir Patil

Work History

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Professional Experience

Research Assistant Scientist

October 2025 to Present

University of Florida, Department of Mechanical & Aerospace Engineering

MAE-A 231, 939 Sweetwater Drive, Gainesville, FL 32611

- Research on mathematical frameworks for compositional control synthesis via Lyapunov methods for nonlinear dynamical systems
- Investigating control approaches utilizing Hamiltonian formulations and differential geometric structures for asymptotic stability analysis
- Co-authoring textbook “Lyapunov-Based Deep Learning for Autonomy” with Springer (anticipated completion May 2026)

Postdoctoral Research Associate

May 2023 to October 2025

University of Florida, Department of Mechanical & Aerospace Engineering

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- Developed Lyapunov-based deep neural network control algorithms with online parameter adaptation using gradient-based weight update laws
- Published 14 peer-reviewed journal papers (including IEEE Trans. Automatic Control, IEEE Trans. Neural Networks and Learning Systems, Automatica) and 7 conference papers during this period
- Implemented MATLAB and Python software packages for Lyapunov-based neural network controllers and projection-based parameter adaptation algorithms

Graduate Research Assistant

September 2019 to May 2023

University of Florida, Nonlinear Controls and Robotics Group

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- Analyzed exponential stability properties of robust integral of the sign of the error (RISE) feedback control using Lyapunov analysis techniques
- Derived Lyapunov-based adaptive parameter update laws for multilayer neural network approximators with arbitrary depth using backpropagation-inspired gradient computations
- Published 3 peer-reviewed journal papers (IEEE Trans. Automatic Control, IEEE Control Systems Letters) and 6 conference papers (IEEE CDC, ACC, IFAC) during this period

Project Associate

July 2018 to April 2019

Indian Institute of Technology Delhi, Department of Electrical Engineering

Hauz Khas, New Delhi, Delhi 110016, India

- Contributed to Tata Consultancy Services (TCS) sponsored project on design, fabrication, modeling and control of aerial manipulator system for pick and delivery applications
- Developed singularity-free hierarchical controller using Barrier Lyapunov Function (BLF) based mathematical modeling and design approach
- Published 2 conference papers (ACC, IFAC) during this period